



Sand solution

When American student **Joseph Woodland** was researching a technology that could identify food products in 1948, he found the solution on a **BEACH**. As he drew lines in the sand with his fingers, he imagined a code of bars and spaces. This resulted in the Universal Product Code (UPC), or bar code system, which was first adopted in 1974 and **continues to be used today**.

Bar code

It revolutionized retail and is the best system for **RECOGNIZING** objects, bar none.

Black and white bars of different widths represent numbers from 1 to 9.

A laser scans the bars for a computer to read.

Product code (last six digits) is unique to the item being sold.

The six-digit manufacturing code identifies the maker of the product.

Guard bars are longer and separate the two codes.

How it changed the world
Bar codes speed up transactions and cut wait times, but they also allow suppliers to track what people are buying.

Code conversion

Bar codes are basically **identity tags**. At first glance, they are a series of vertical bars of different widths on a product's packaging. When read by a **LASER SCANNER**, the item is quickly identified, along with information about it, such as price and stock information. About **five billion** bar codes are scanned every day.

